EXP NO:18 EXPERIMENT TO UNDERSTAND KNN ALGORITHM FOR A GIVEN DATASET

CODE :

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/Iris_Dataset.csv')
data.info()
df.variety.value_counts()
df.head()
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,ra
ndo
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
```

```
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report

print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT :

<class 'pandas.core.frame.DataFrame'> RangeIndex: 150 entries, 0 to 149 Data columns (total 5 columns): # Column Non-Null Count Dtype --- ------ -------------- ----- 0 sepal.length 150 non-null float64 1 sepal.width 150 non-null float64 2 petal.length 150 non-null float64 3 petal.width 150 non-null float64 4 variety 150 non-null object dtypes: float64(4), object(1) memory usage: 6.0+ KB None variety Setosa 50 Versicolor 50 Virginica 50 Name: count, dtype: int64 sepal.length sepal.width petal.length petal.width variety 0 5.1 3.5 1.4 0.2 Setosa 1 4.9 3.0 1.4 0.2 Setosa 2 4.7 3.2 1.3 0.2 Setosa 3 4.6 3.1 1.5 0.2 Setosa 4 5.0 3.6 1.4 0.2 Setosa 0.9666666666666667 1.0 [[50 0 0] [0 47 3] [0 1 49]] precision recall f1-score support Setosa 1.00 1.00 1.00 50 Versicolor 0.98 0.94 0.96 50 Virginica 0.94 0.98 0.96 50 accuracy 0.97 150 macro avg 0.97 0.97 0.97 150 weighted avg 0.97 0.97 0.97 150